

INDOORPULSE — PROJECT MILESTONE PLAN

Course: UCS503 (Software Engineering Project)

Project: IndoorPulse — Infrastructure-Free Indoor Positioning System

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Project Timeline Overview

- **Milestone 1 (Weeks 1–3):** System Architecture & Cloud Portal — **Status: Completed**
- **Milestone 2 (Weeks 4–6):** Core Sensor & Step Detection Engine — **Status: Completed**
- **Milestone 3 (Week 7):** Mid-Term Prototype Presentation (Sept 14–18) — **Status: Current**
- **Milestone 4 (Weeks 8–10):** Floor Mapping & Graph Routing — **Status: In Progress**
- **Milestone 5 (Weeks 11–13):** Particle Filter & Drift Correction — **Status: Planned**
- **Milestone 6 (Weeks 14–16):** Field Trials & Final Evaluation — **Status: Planned**

Detailed Milestones

Milestone 1: Setup & Architecture (Weeks 1–3) — Completed

- **Goal:** Establish system requirements, development workflows, and the cloud presentation portal.
- **Work Completed:**
 - Drafted system requirements, initial architecture, and component diagrams.
 - Initialized the native Android codebase using Kotlin and Jetpack Compose.
 - Deployed the public project website via GitHub Pages.
 - Provisioned an AWS EC2 instance running Flask and Nginx with SSL for deliverable hosting.

- Integrated Backblaze B2 object storage and SQLite metadata tracking for version control.
- **Deliverable:** Live project portal and initialized code repository.

Milestone 2: Sensor Fusion & Step Engine (Weeks 4–6) — Completed

- **Goal:** Build and calibrate the client-side Pedestrian Dead Reckoning (PDR) engine on Android.
- **Work Completed:**
 - Configured Android SensorManager to sample linear acceleration and rotation vectors at 50 Hz.
 - Implemented a dual-threshold Schmitt trigger to classify steps and eliminate hand-shake noise.
 - Added dynamic stride estimation based on step cadence and vertical acceleration amplitude.
 - Built stationary zero-velocity checks (ZUPT) to prevent position drift when not walking.
- **Deliverable:** Calibrated step-detection and distance-tracking engine running on real hardware.

Milestone 3: Working Prototype Presentation (Week 7: Sept 14–18) — Current Milestone

- **Goal:** Demonstrate the working prototype and technical integration during scheduled lab evaluations.
- **Active Tasks:**
 - Connect the step engine to a custom Jetpack Compose canvas rendering a live position puck and orientation chevron.
 - Publish verified system documentation (4 Use Case, 5 Sequence, 1 Detailed Class Diagram) to the cloud repository.
 - Conduct a live, end-to-end corridor traversal demonstration proving offline tracking without GPS.

- **Deliverable:** Functional Android APK demo and integrated 3-tier presentation portal.

Milestone 4: Graph Mapping & Routing (Weeks 8–10) — In Progress

- **Goal:** Implement localized floor-plan mapping and graph-based shortest-path routing.
- **Planned Tasks:**
 - Build an in-app "Surveyor Mode" enabling users to plot junction nodes and walkable corridor edges.
 - Configure an offline Room (SQLite) database to store and query spatial graph entities.
 - Implement Dijkstra's algorithm to calculate the shortest path between selected start and end points.
 - Optimize graph traversal by implementing the A* search algorithm for larger floor plans.
- **Deliverable:** Autonomous, offline pathfinding engine running on local map databases.

Milestone 5: Drift Correction & Particle Filter (Weeks 11–13) — Planned

- **Goal:** Bound cumulative sensor drift over long-distance walking paths.
- **Planned Tasks:**
 - Implement a 2D Monte Carlo particle filter running on background coroutine threads.
 - Apply hard topological wall boundaries to prevent estimated positions from clipping through walls.
 - Calibrate compass orientation transforms to align physical walking headings with map coordinates.
 - Introduce Wi-Fi RSSI reference anchors at major junctions for automatic drift resets.
- **Deliverable:** Stabilized tracking maintaining under 5% cumulative lateral drift over 50 meters.

Milestone 6: Multi-Floor & Final Evaluation (Weeks 14–16) — Planned

- **Goal:** Complete multi-floor navigation logic, extensive field testing, and project defense.
- **Planned Tasks:**
 - Add manual floor-toggle switches and vertical transition nodes (stairwells and elevators).
 - Execute 100-meter closed-loop walking benchmarks to validate return-point accuracy.
 - Profile and optimize battery consumption during extended background sensor polling.
 - Assemble the final project report, user documentation, and Git commit audit trail.
- **Deliverable:** Production-ready Android APK, comprehensive report, and final project defense.

Measurable Target Metrics

- **Step Detection Accuracy:** $\geq 95\%$ across normal walking speeds (1.2 Hz – 2.0 Hz).
- **False Step Rejection:** 0 false-positive steps detected during stationary handling or surface vibration.
- **Route Calculation Latency:** ≤ 20 ms on topological graphs containing up to 500 nodes.
- **Corridor Tracking Drift:** $\leq 5\%$ lateral error over a continuous 50-meter walking segment.
- **Offline Independence:** 100% functionality of all core navigation features with zero cellular or Wi-Fi connectivity.